

In this document the fractions includes the floor operation on the result of division. ## Part 1:

1)

If W is a prefix of S the computation of F(W) with only F(S) and |W| is impossible because:

$$F(W) = \frac{F(S)}{131^{|S|-|W|}}$$

2)

$$F(S) = F(W) * 131^{|W|} + F(Z)$$

$$F(W) = \frac{F(S) - F(Z)}{131^{|Z|}}$$

$$F(Z) = F(S) - F(W)$$

Part 2:

DS = HashMap -> key, value = (F(S_i), number of occorence)

hash function H(F(S_i)) = F(S_i) % number

Time complexity: - Insert/Search element = O(1) - Building = O(|D|) = O(M) -
Computation of the number of viruses found = O(|D|) = O(M) Space complexity
= O(|D|) = O(M)